ParSe v2: predict phase-separating protein regions from the primary sequence

This is a form that allows you to enter a set of sequences and quickly find those predicted to drive phase separation. The algorithm also determines if the user-provided set is enriched or depleted in phase separation potential compared to the human proteome. Output includes a FASTA file containing the predicted phase-separating regions.

To analyze a single protein sequence using ParSe v2, please visit our other site.

The algorithm is described in Ibrahim et al, *J. Biol. Chem.* **299**, 102801 (2023).

This web application and its output is described in Wilson et al, *Protein Science*, doi: 10.1002/pro.4756.

Input File

Input a FASTA file containing protein sequences, preferably in the UniProtKB format. Files with tens of thousands of sequences may take a minute or longer to analyze. The algorithm ignores sequences longer than 10,000 residues or shorter than 25 residues.

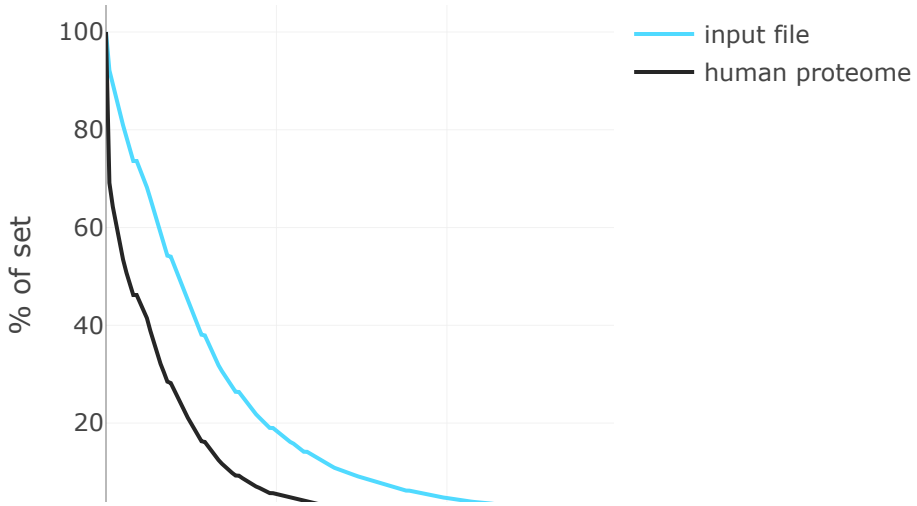
File: Seleccionar archivo proteoma_ncrassa.fasta

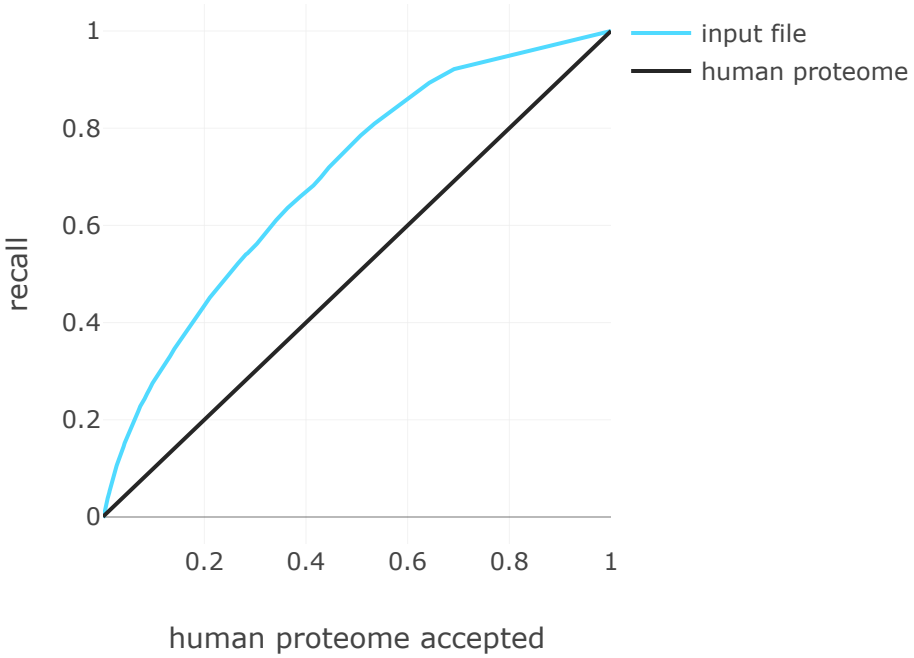
Name for the file (optional):
NCRASSA

Submit

A

Recall plot AUC = 0.6963

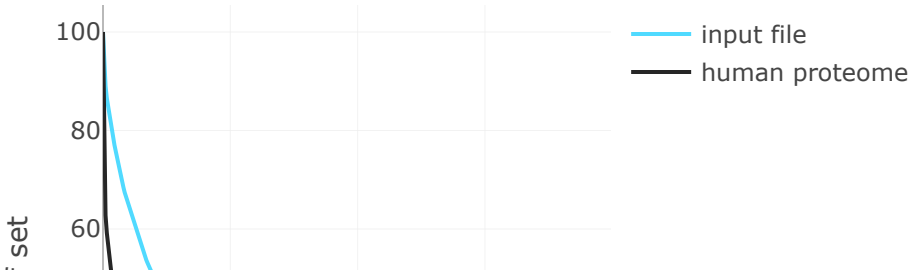


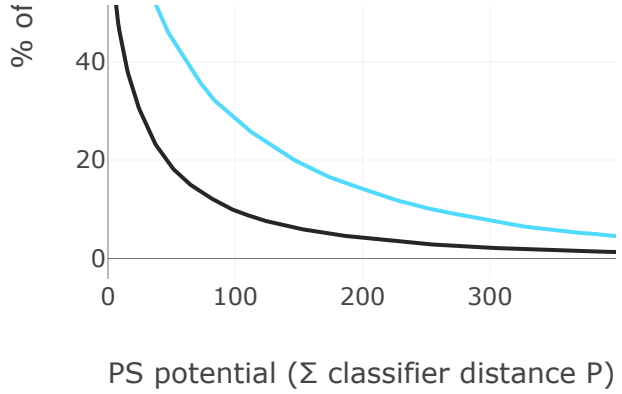


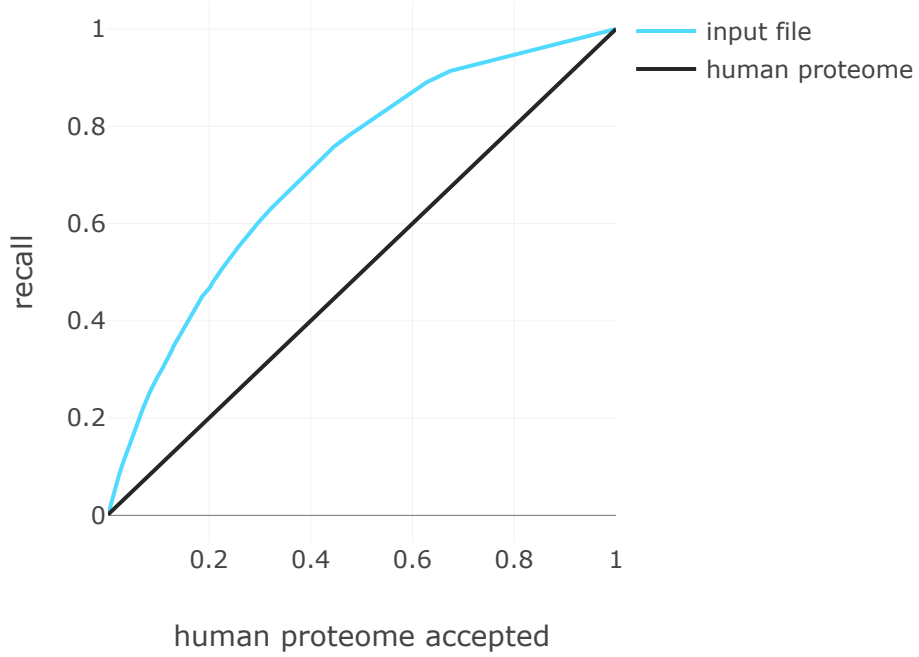
A. Distribution of predicted PS IDR lengths. ParSe v2 finds regions within proteins that are >90% labeled P, which are predicted to be PS IDRs. The y-axis is the percent of proteins in a set with PS IDRs at least as long as the length indicated by the x-axis. The recall plot compares the percent of set for the human proteome against itself and against the input file.

B

Recall plot AUC = 0.7116







B. Distribution of PS potentials. ParSe v2 calculates the classifier distance sum of windows labeled P, which we use as a numerical score to estimate the PS potential. The y-axis is the percent of proteins in a set with a PS potential equal to or greater than the value indicated by the x-axis. The recall plot compares the percent of set for the human proteome against itself and against the input file.

Sequence-calculated properties (sort this table by its 3rd or 4th column to find phase-separating proteins):

Download CSV

Seq…	Seq…	Σ cl…	leng…	first…	last …	Σ cl…	leng…	first…	last …

1	473	1.3	0	0	0	63.3	0	0	0
2	250	0.0	0	0	0	0.0	0	0	0
3	685	66.4	0	0	0	90.2	29	420	448
4	762	52.4	0	0	0	76.5	32	227	258
5	557	10.7	0	0	0	0.0	0	0	0
6	619	185.0	30	36	65	11.1	0	0	0
7	231	16.1	0	0	0	46.9	0	0	0

Plot A percent of set and recall data: Download CSV

length of longest PS IDR	count	% of set, input file	% of set, hum
0	10250	100	100
1	9446	92.15609756097561	69.1317164
2	9157	89.33658536585367	64.2556612
3	8844	86.28292682926829	60.1916119
4	8559	83.50243902439024	56.5054267
5	8296	80.93658536585366	53.4704542
6	8045	78.48780487804878	50.7503684
7	7795	76.04878048780488	48.3317700
8	7545	73.60975609756098	46.1918799
9	7545	73.60975609756098	46.1918799
10	7373	71.93170731707316	44.5062307

Plot B percent of set and recall data: Download CSV

PS potential (Σ classifier di…	count	% of set, input file	% of set, hum
0	10250	100	100
1	9366	91.37560975609756	67.4567868
2	9127	89.0439024390244	62.7870829
3	8873	86.56585365853658	59.2174728
4	8675	84.6341463414634	56.3138148
5	8491	82.83902439024389	53.8831569
6	8326	81.22926829268293	51.6414310
7	8166	79.66829268292683	49.6904730
8	8027	78.31219512195122	47.7770333
9	7898	77.05365853658537	46.1423020
10	7770	75.8048780487805	44.5611684

Predicted PS IDRs (from input file, those w/ $N \geq 50$): Download FASTA